Module 02 • System Architecture

IRRIGATEWISE System Architecture & Engineering Blueprint

IRRIGATEWISE Hydrogel-Based Smart Irrigation System Specification

IRRIGATEWISE System Architecture & Engineering Blueprint

1. Executive Technical Overview

IRRIGATEWISE is an intelligent, closed-loop agricultural engineering system designed to optimize irrigation water consumption through hydrogel polymer soil amendment and predictive machine learning.

Traditional precision irrigation architectures rely on static threshold-based rule sets (such as Soil Moisture Deficit / SMD calculation). While effective at preventing acute plant wilting, threshold mechanisms suffer from **reactive hysteresis**: irrigation is triggered only after the root zone has already depleted to critical thresholds, and cannot account for upcoming precipitation or diurnal evapotranspiration peaks. Furthermore, conventional systems do not account for the moisture buffering kinetics introduced by Superabsorbent Polymers (Hydrogel).

IRRIGATEWISE implements an end-to-end **SENSE** → **PREDICT** → **DECIDE** → **ACT** → **VERIFY** autonomous feedback control loop:

1. SENSE (Physical & Simulated Edge)

Capacitive Moisture Sensor (v1.2) + DHT22 (Ambient Temp & Humidity) + Rain Gauge → **ESP32 Microcontroller Node**

↓ JSON Telemetry over HTTP/Wi-Fi

2. INGESTION & ANOMALY SCREENING

POST /api/v1/sensors/readings → **Isolation Forest & Physical Sanity Filter** → Valid Telemetry into PostgreSQL (Outliers quarantined to anomaly_events)

↓ Validated Historical Window

3. PREDICT (Machine Learning Inference)

Feature Pipeline (Lags, Rolling Means, Cyclical Encodings) → **Gradient Boosting Multi-Horizon Regressor (1h, 3h, 6h)** → Forecasts logged to ml_predictions

↓ Predictive Forecast at t+3h

4. DECIDE (Dual-Mode Decision Engine)

Mode 1: Agronomic SMD Heuristic | Mode 2: Autonomous ML + Hydrogel Offset (Rain Forecast ≥ 5mm triggers 100% water suppression)

↓ Final Irrigation Decision Matrix

5. ACT (Pump Controller & Hardware Interlocks)

Safety Interlocks: Cooldown Hold (30 min) + Saturation Ceiling (≥85%) + Max Duration Clamp (25 min) → Submersible 12V Drip Pump Relay → Logged to irrigation_events

↓ Real-Time Feedback Telemetry

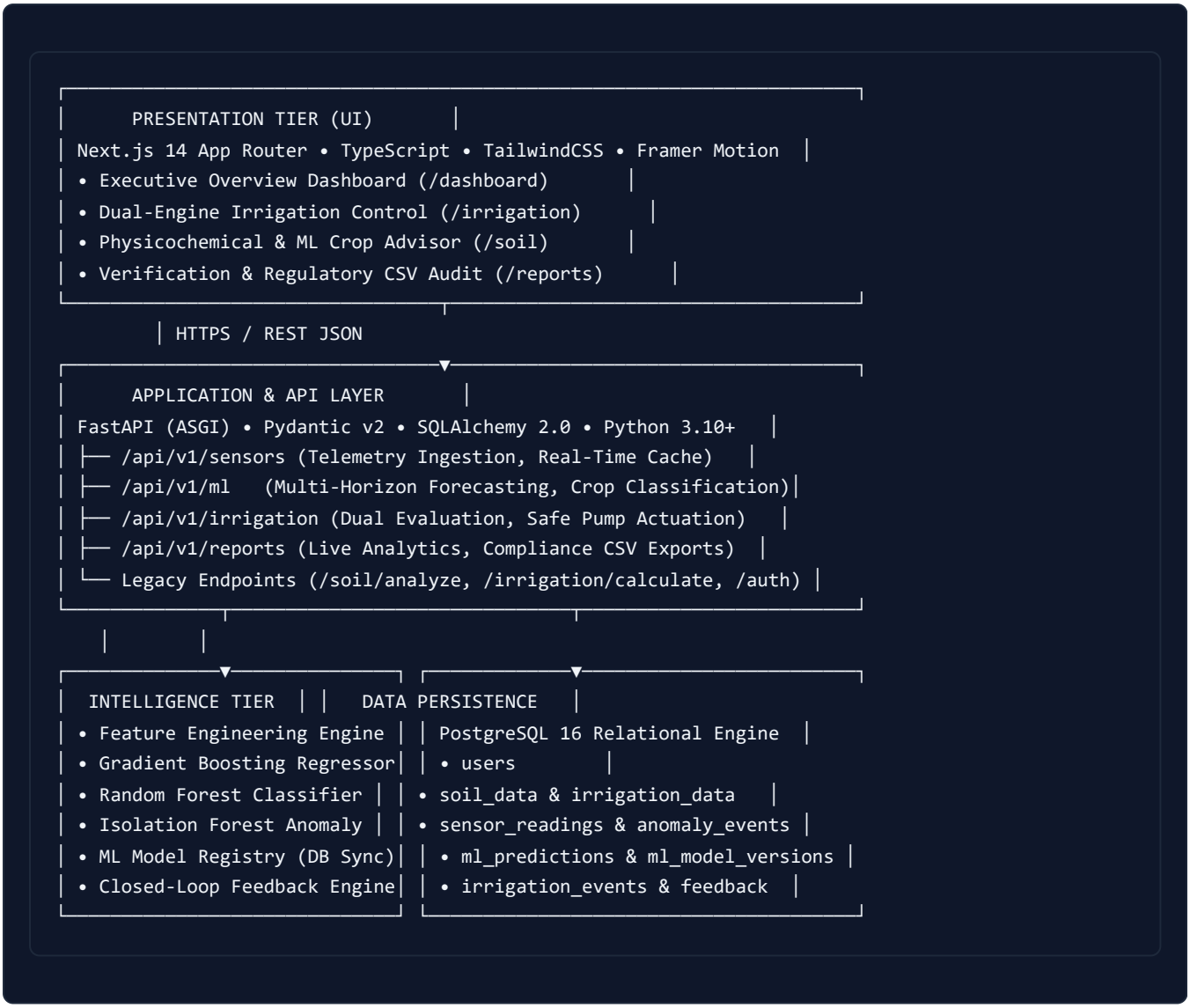
6. VERIFY (Closed-Loop Telemetry Auditing)

Ground Truth Verification at t+1h, t+3h, t+6h → **Model Drift Auditing (MAE / RMSE Tracking)** → Logged to prediction_feedback

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2. Multi-Tier System Topology

The IRRIGATEWISE infrastructure is organized into four decoupled tiers:



3. Data Flow and Ingestion Pipeline

Step 1: Telemetry Acquisition

- Edge microcontrollers (ESP32) sample capacitive soil moisture, temperature, humidity, and rainfall every 60 seconds.
- Payloads are assembled into RFC 8259 JSON objects containing node identifier, plot type (`hydrogel` or `control`), and sensor readings.

Step 2: Quality Filtering & Anomaly Screening

- Upon reception at `/api/v1/sensors/readings` , the payload is screened by `backend/ml/anomaly_detection.py` .
- **Physical Out-of-Bounds:** SM ∉ [5%, 95%], Temp ∉ [0°C, 60°C].
- **Spike Detection:** Sudden jumps >15% in under 5 minutes without pump actuation.
- **Freeze Detection:** Sensor values invariant over 6 hours (indicating sensor detachment or ADC failure).

- **Decoupling Detection:** Pump reported `ON` for >20 minutes without positive moisture rate of change (dSM/dt ≤ 0).

Step 3: Feature Synthesis & Multi-Horizon Inference

- The feature pipeline computes rolling statistics (3h rolling mean, 6h rolling mean, rolling standard deviation), time lags (SM_t-1, SM_t-2, SM_t-3), cyclical diurnal features (sin(2π h/24), cos(2π h/24)), and hydrogel interaction terms.
- The pre-trained Multi-Output Regressor outputs soil moisture forecasts for t+1h, t+3h, and t+6h.

Step 4: Closed-Loop Actuation with Physical Safety

- The decision engine evaluates if SM_3h drops below the crop management allowed depletion (MAD).
- If true, water depth requirement (mm) and volume (L) are computed, factoring in hydrogel retention efficiency.
- Safety controller verifies: 1. Cooldown interlock (minimum 30 minutes since last pump event). 2. Maximum runtime interlock (hard timeout at 25 minutes). 3. Moisture saturation ceiling (immediate abort if moisture ≥ 85%). 4. Global emergency stop switch.

Step 5: Post-Hoc Truth Verification

- Once elapsed real time reaches t+1h, t+3h, t+6h, the system compares predicted moisture with true observed telemetry, writing records to `prediction_feedback` to dynamically track Mean Absolute Error (MAE) and model degradation.

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4. Hardware and Communication Interfaces

Component	Hardware Specification	Interface Protocol	Role in System
Microcontroller	ESP32-WROOM-32D Dual-Core 240MHz	Wi-Fi 802.11 b/g/n, HTTP POST	Edge telemetry collection & relay driving
Soil Sensor	Analog Capacitive Moisture Sensor v1.2	ADC (GPIO 34, 12-bit)	Corrosion-resistant dielectric moisture sensing
Ambient Sensor	DHT22 (AM2302)	1-Wire Digital (GPIO 4)	Air temperature & relative humidity
Rain Sensor	Optical / Conductive Digital Rain Detector	Digital Input (GPIO 18)	Precipitation detection for rain suppression
Pump Actuator	5V Relay Module controlling 12V DC Pump	Digital Output (GPIO 23)	Closed-loop precision drip irrigation

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5. Security and Operational Integrity

1. **Authentication:** Stateless JSON Web Token (JWT) architecture using HMAC-SHA256 signatures with 24-hour expiration tokens.
2. **Password Cryptography:** High-entropy salt and password hashing via `bcrypt`.
3. **Data Protection:** Parameterized queries mediated by SQLAlchemy ORM to eradicate SQL injection vulnerabilities.
4. **Resilience & Fallback:** In the event of network disruption or ML service latency, the system automatically falls back to deterministic Rule-Based Agronomic SMD calculations to ensure zero crop desiccation risk.